

Kapitza Pendulum Project

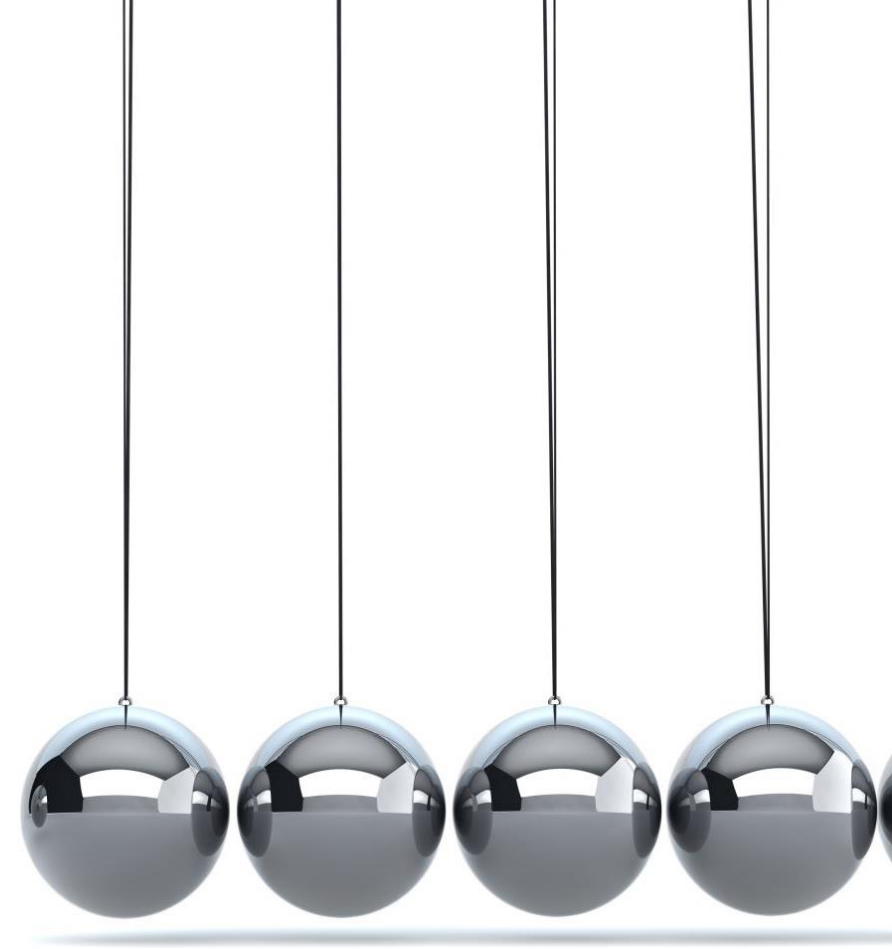


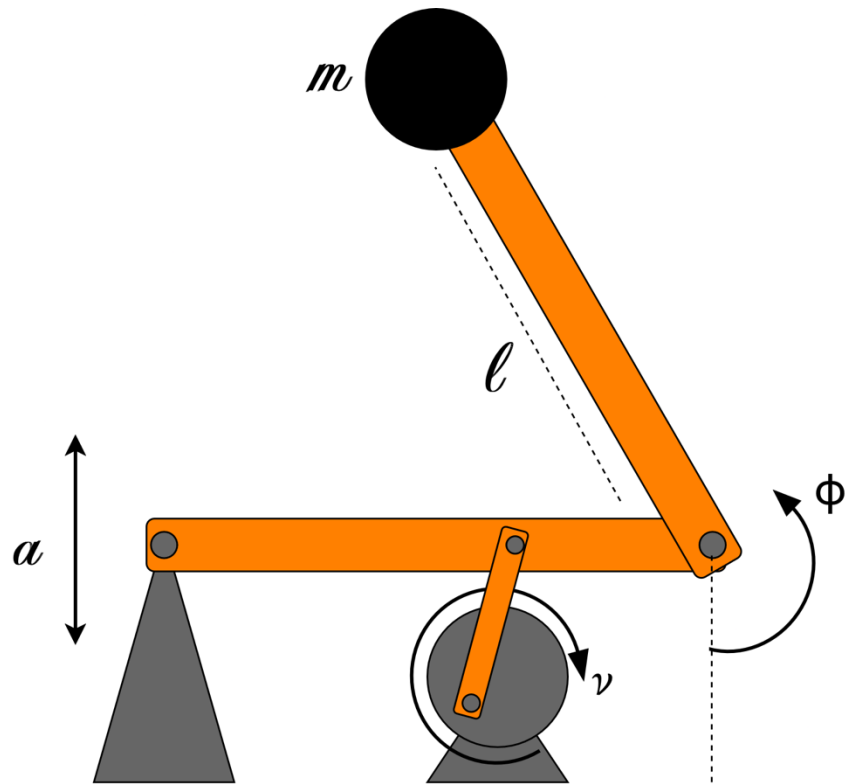
NEIL GHUGARE

PHYSICS 5300 – AUTUMN 2025

History

- Rigid pendulum with vertically-vibrating pivot point.
- 1951: Russian Nobel laureate Pyotr Kapitza.
- Multiple interesting and unusual properties.

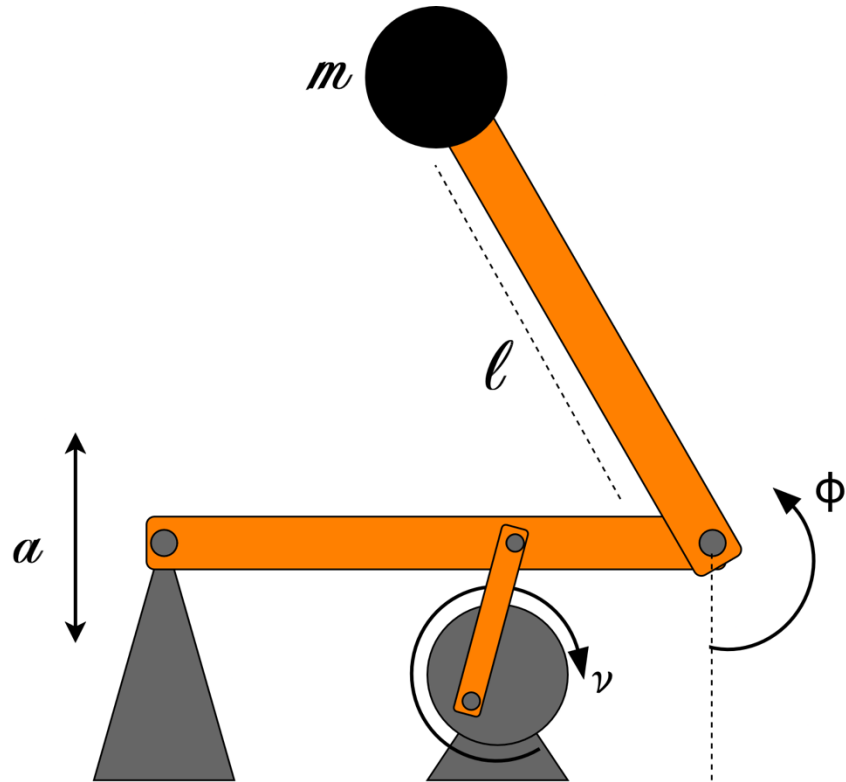




The System

- a : The driving amplitude.
- ℓ : The pendulum length.
- ν : The driving frequency.
- m : The mass of the bob.
- ϕ : The angle of the pendulum, with 0 being downwards, and increasing counterclockwise.
- Assumptions: No drag, no dissipative forces, linear deflection.

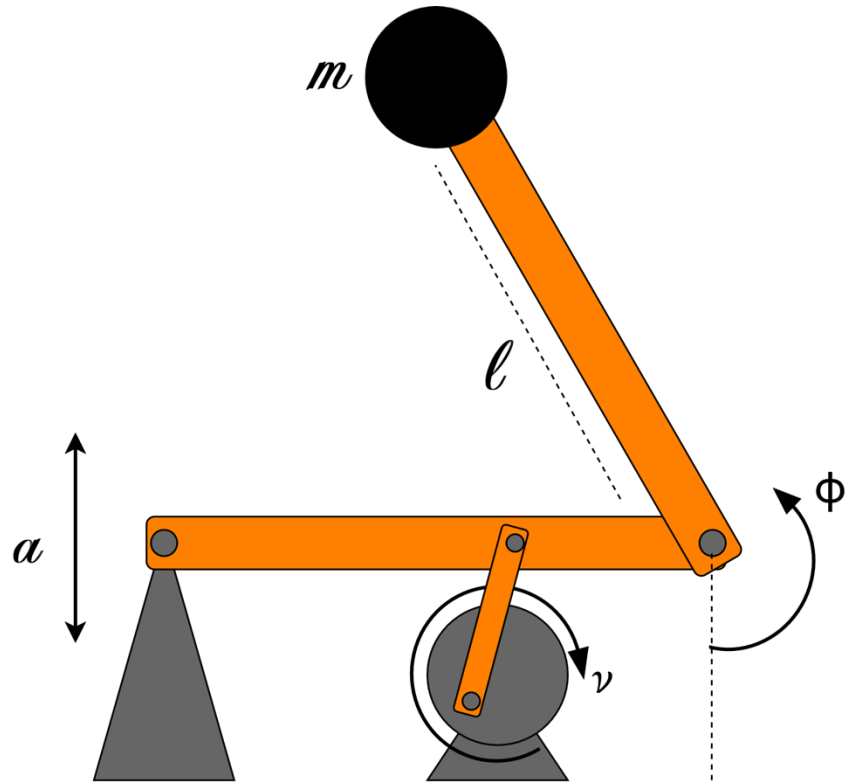
Lagrangian and EoM



$$\mathcal{L} = \frac{1}{2} m \ell^2 \dot{\phi}^2 + m a \ell v \dot{\phi} \sin(\phi) \sin(vt) + m g \ell \cos(\phi)$$

$$\ddot{\phi} = -(g + a v^2 \cos(vt)) \frac{\sin(\phi)}{\ell}$$

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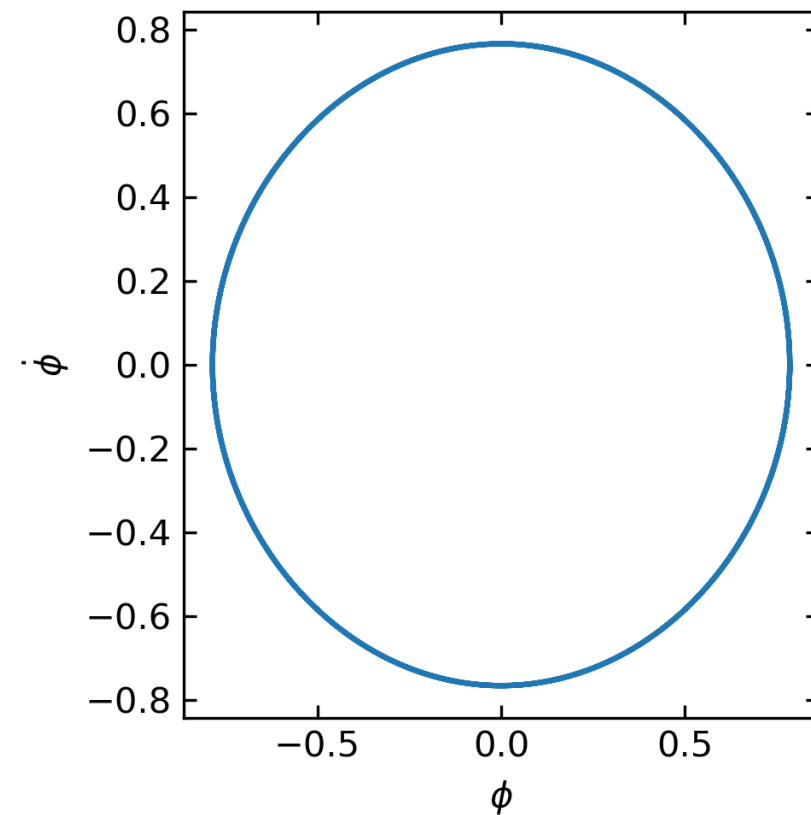
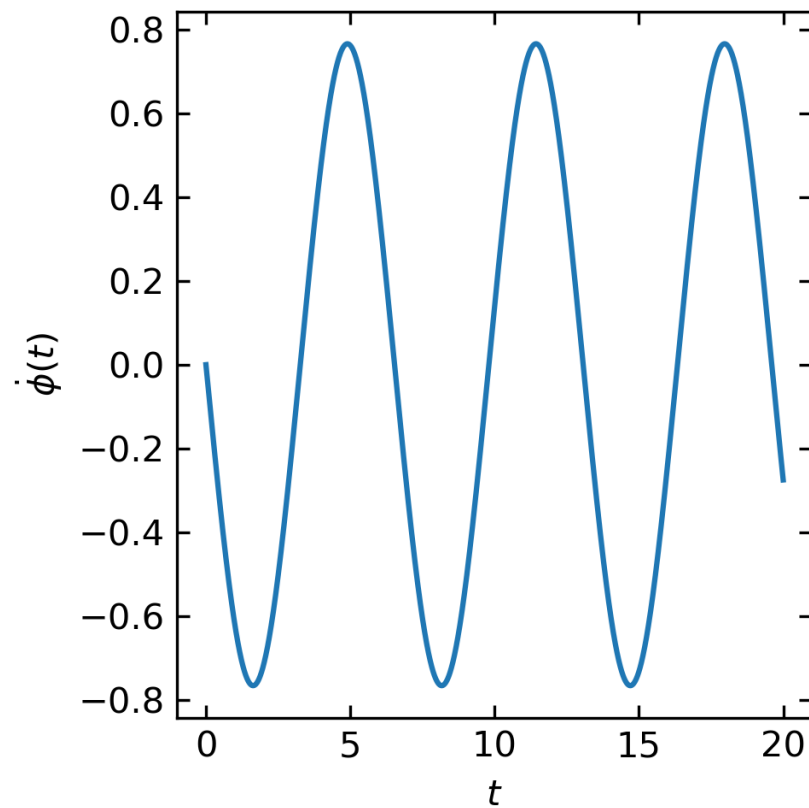
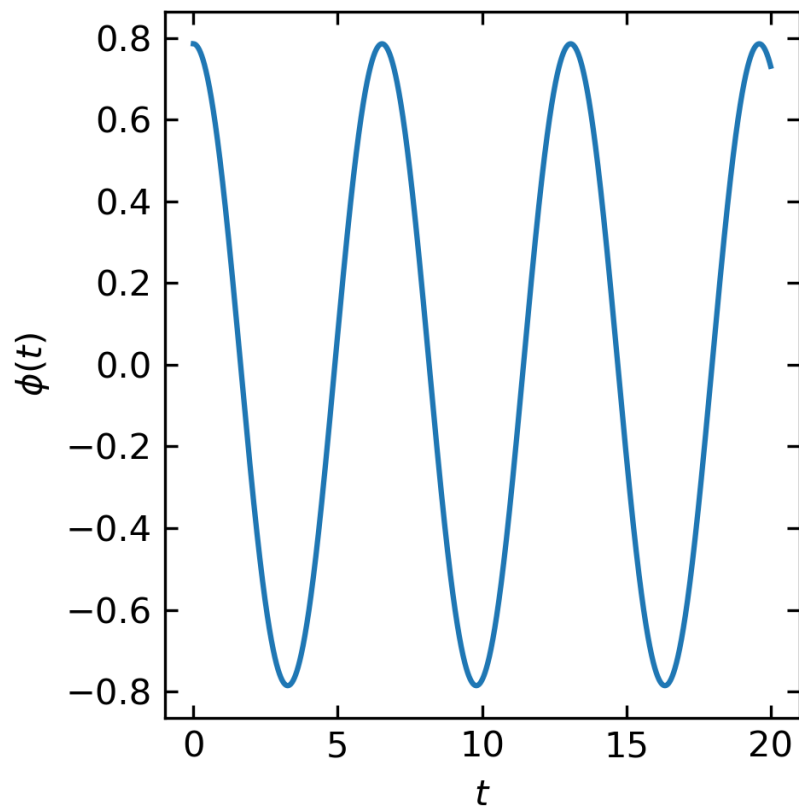
$$\ddot{\phi} = -(g + a v^2 \cos(vt)) \frac{\sin(\phi)}{\ell}$$

$$\ddot{\phi} = -\omega_0^2 (1 + \epsilon f^2 \cos(f \omega_0 t)) \sin(\phi)$$

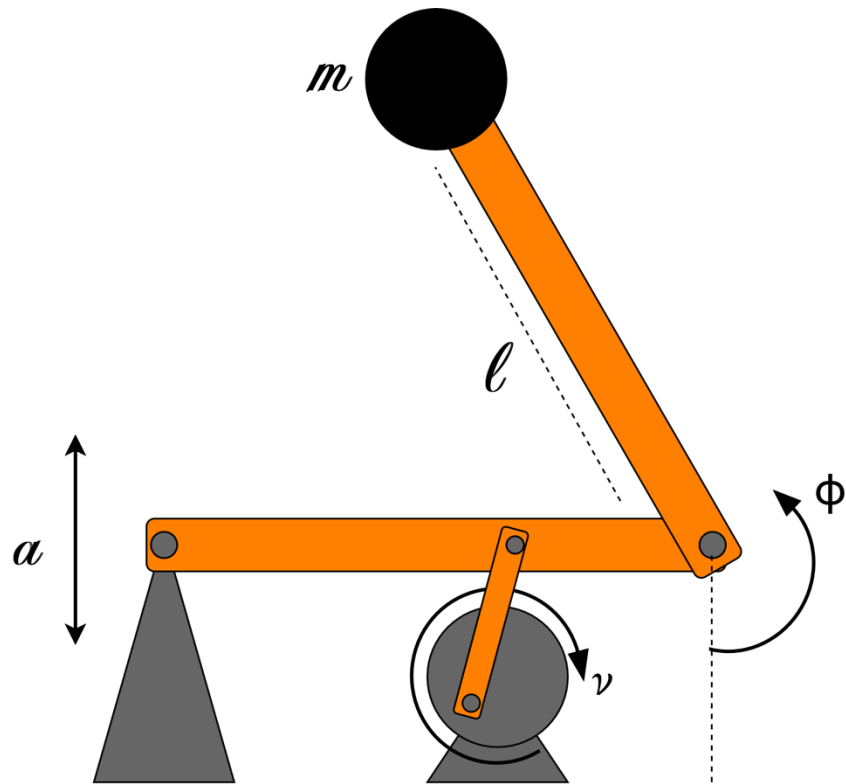
$$\epsilon \equiv a / \ell, f \equiv v / \omega_0$$

Motion and State Space for Kapitza Pendulum

$$\omega_0 = 1.0, a = 0, \nu = 0$$



High-Driving, Low Amplitude Regime

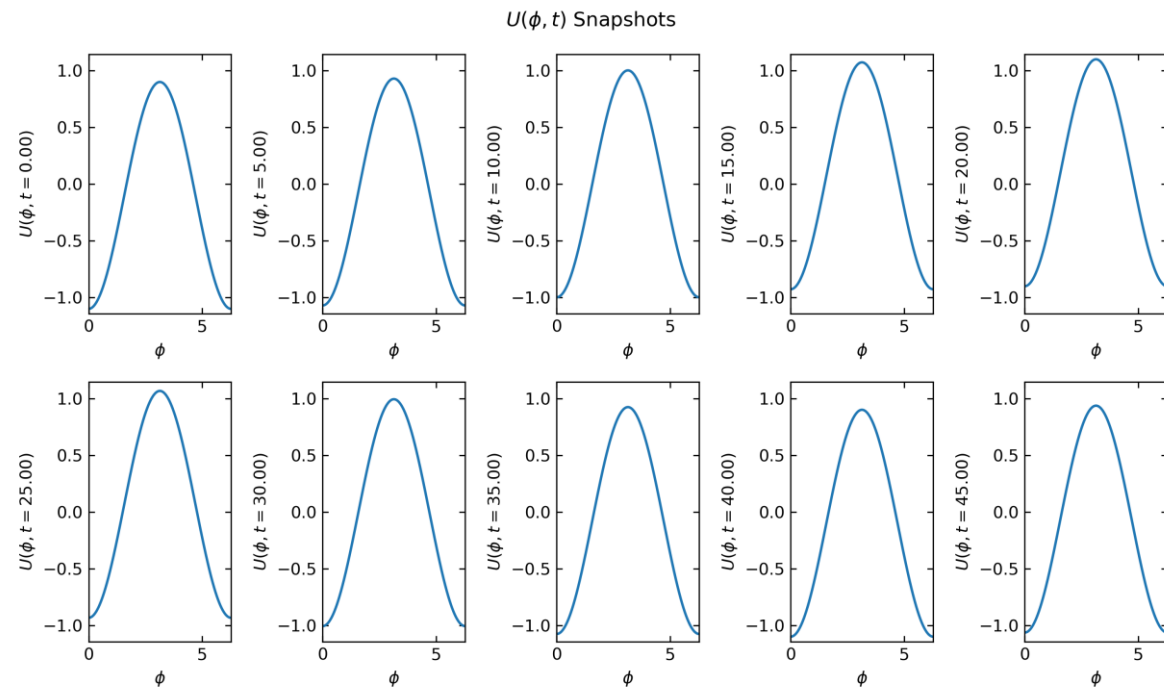


$$\ddot{\phi} = -\omega_0^2(1 + \epsilon f^2 \cos(f\omega_0 t)) \sin(\phi)$$

$$\epsilon \equiv a/\ell, f \equiv v/\omega_0$$

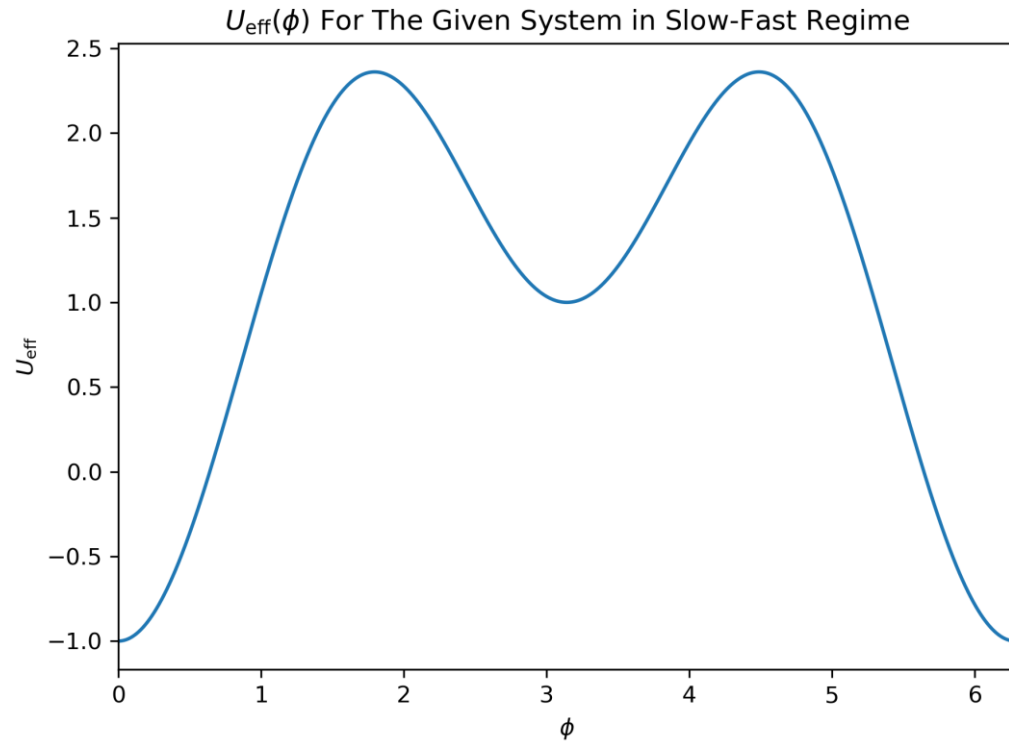
$$\epsilon \ll 1, f \gg 1$$

$$U(\phi, t) = -mg(l \cos(\phi) + a \cos(\nu t))$$



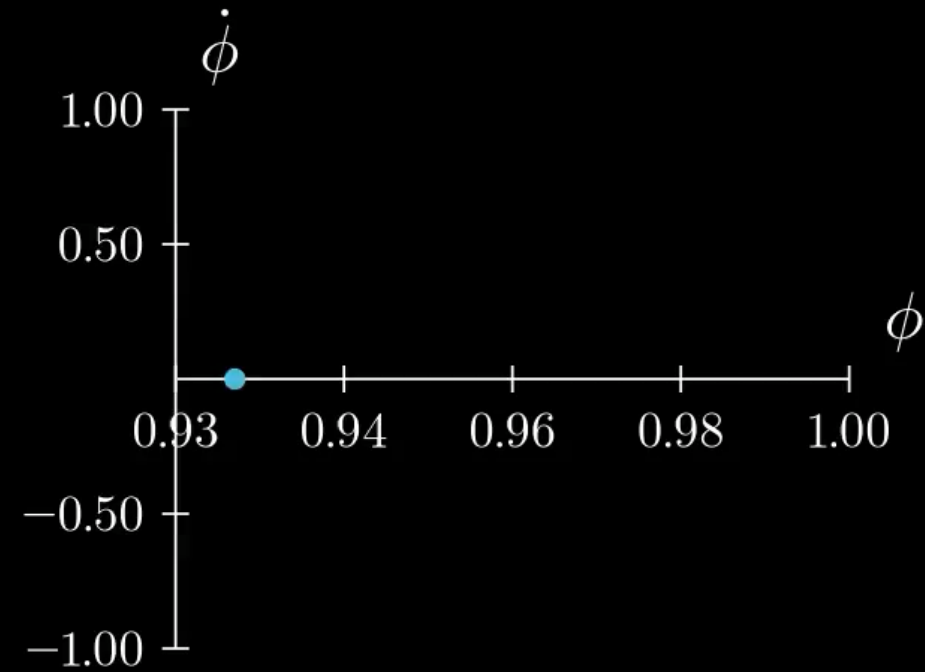
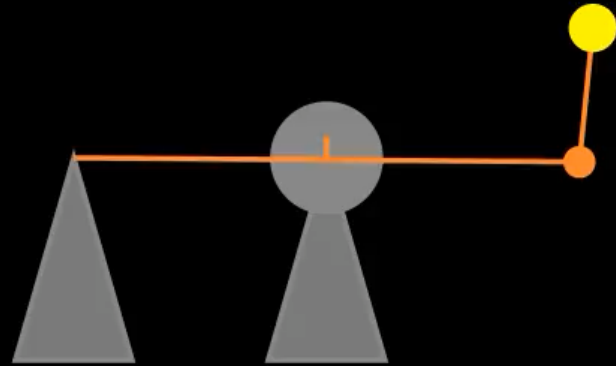
- $a = 0.1$
- $\nu = 30$
- $m = 1$
- $g = 1$
- $l = 1$

$$U_{eff}(\phi) \approx -mgl \cos(\phi) + \frac{ma^2v^2}{4} \sin^2(\phi)$$



- Effective potential for the high-driving, low-amplitude regime.
- Derivation part of project.
 - Fast-slow decomposition of motion.

Kapitza Pendulum: $\omega_0 = 1.00$, $\epsilon = 0.10$, $f = 30.00$, $\phi_0 = 3.04$, $\dot{\phi}_0 = 0.00$
 $t = 0.00\text{s}$



(Normalized, ϕ may not be centered at 0.)
($\min \phi = 3.02$, $\max \phi = 3.26$, $\min \dot{\phi} = -0.39$, $\max \dot{\phi} = 0.39$)

How Does the Amplitude Impact Stability?



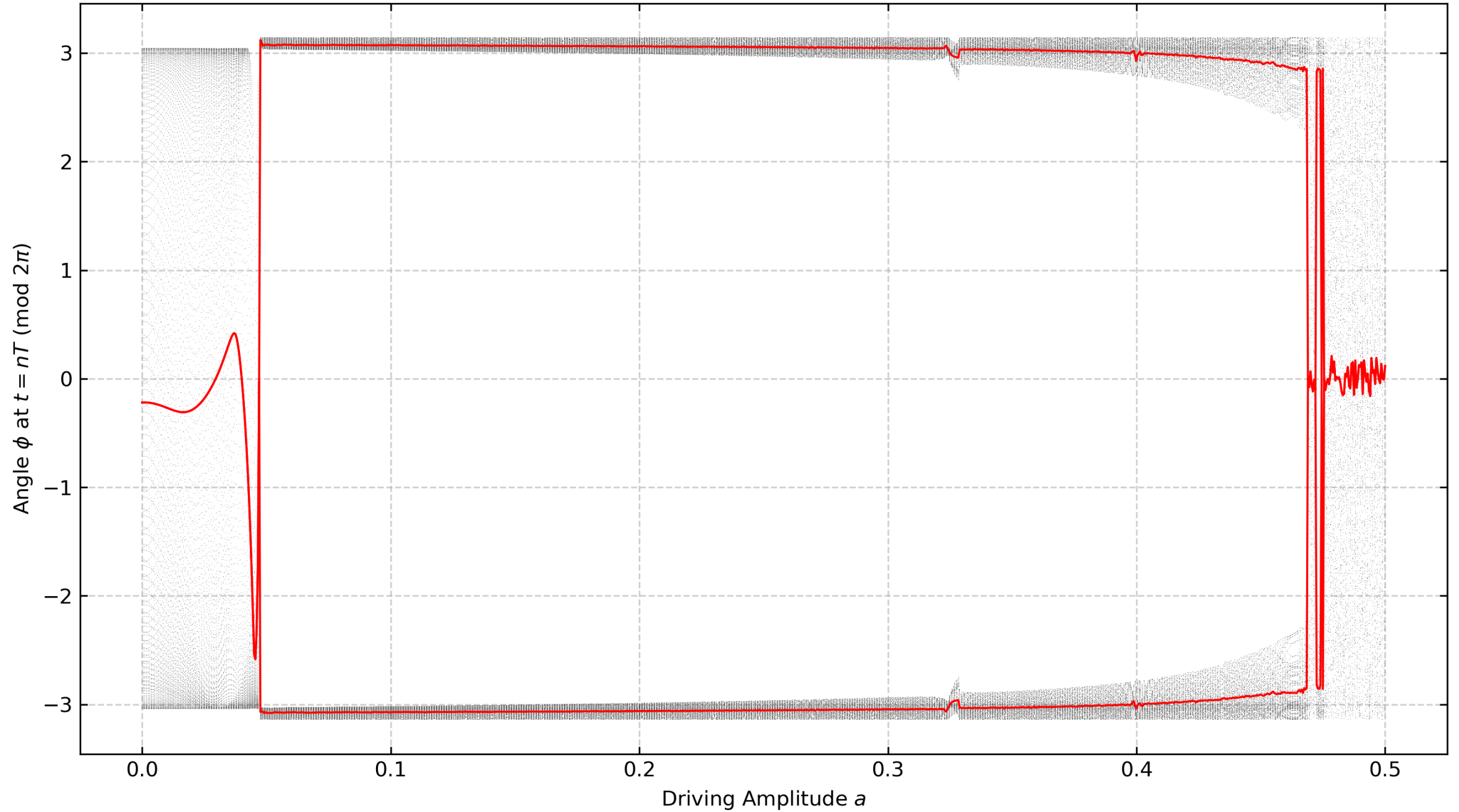
CHAOS AND BIFURCATION ANALYSIS OF THE
SYSTEM.



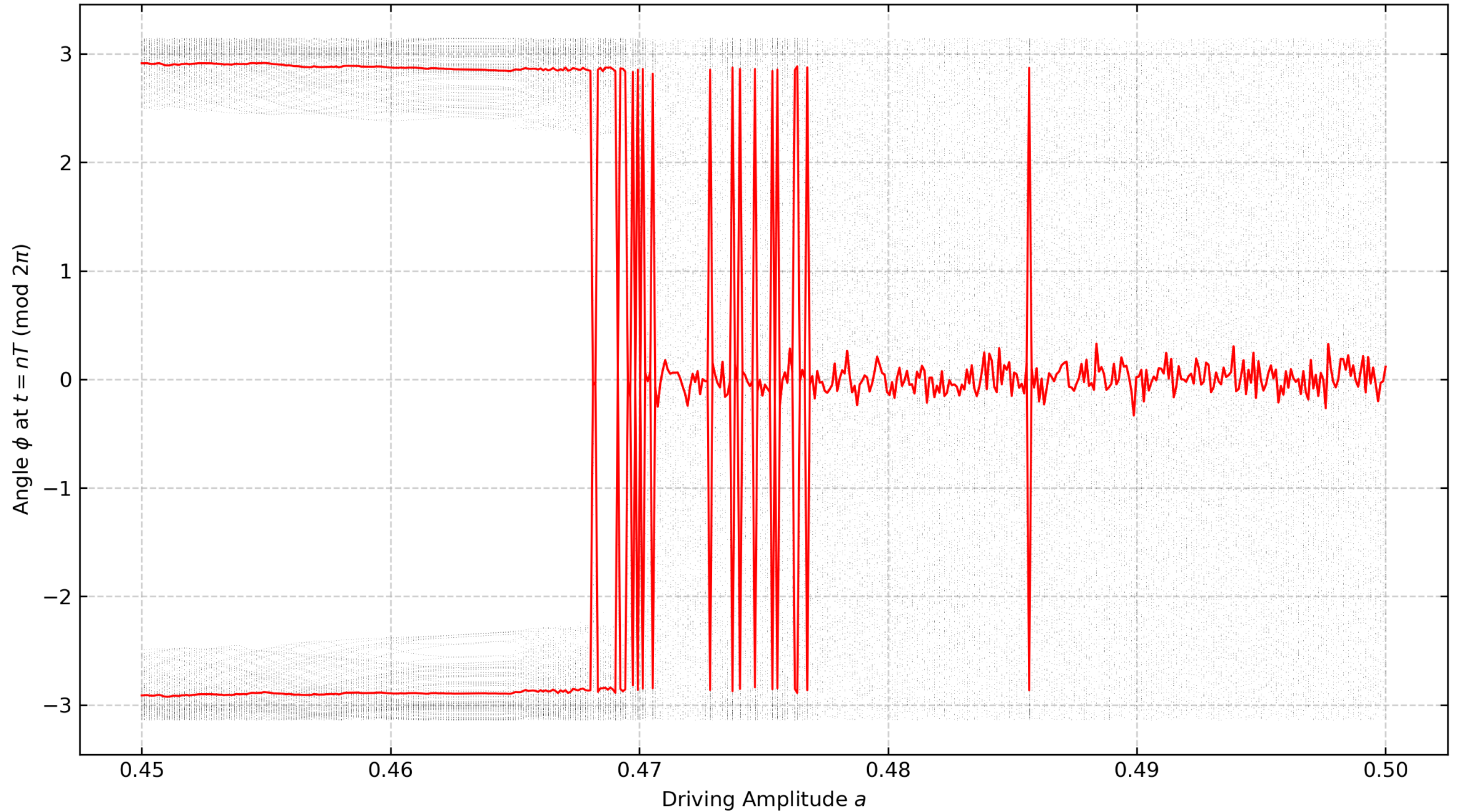
How Do We Tackle This?

- Hopf bifurcation.
 - Continuous scale $\rightarrow$ discrete scale.
- Liapunov exponent.
- Fix the driving frequency, $\nu = 30$

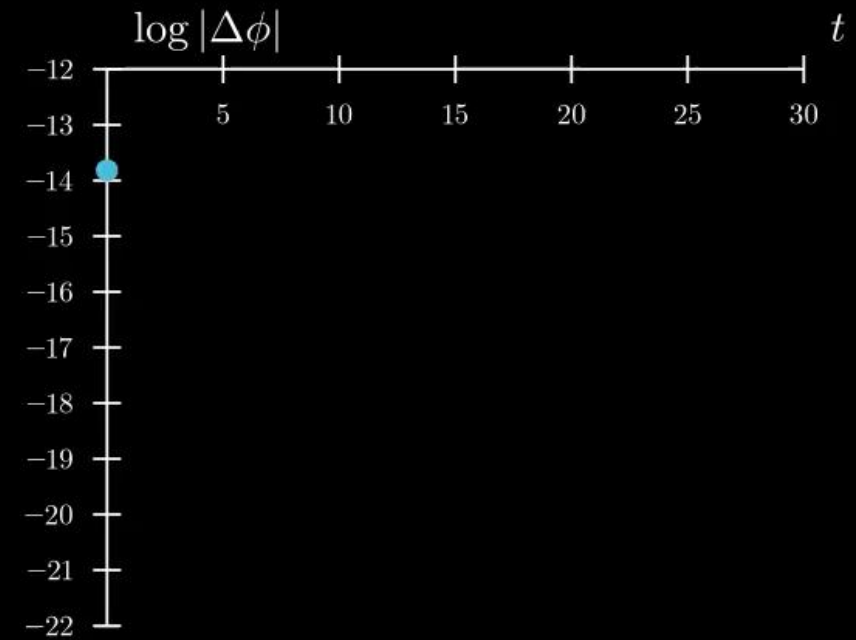
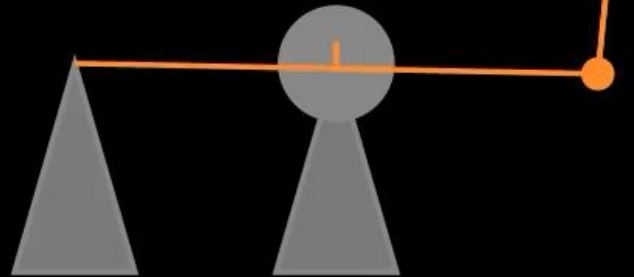
Bifurcation Diagram for Kapitza Pendulum ($\nu = 30.0$)



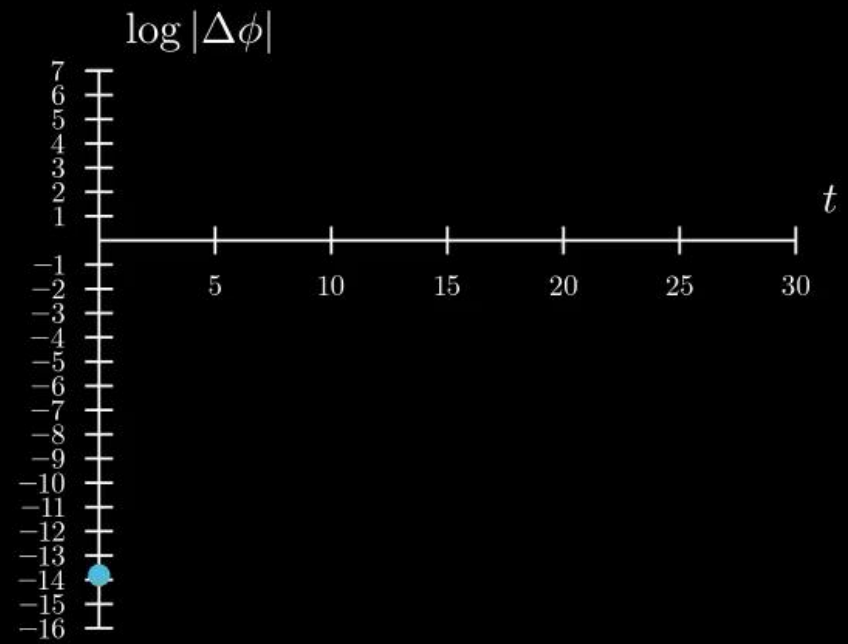
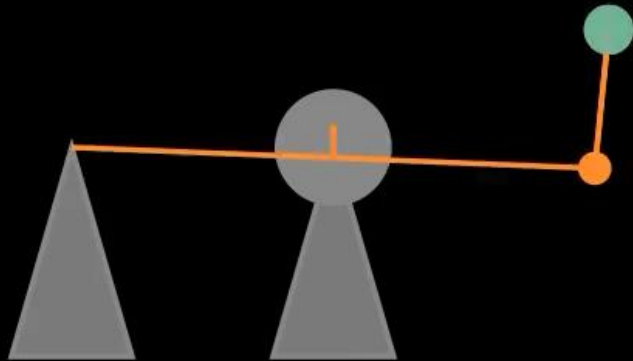
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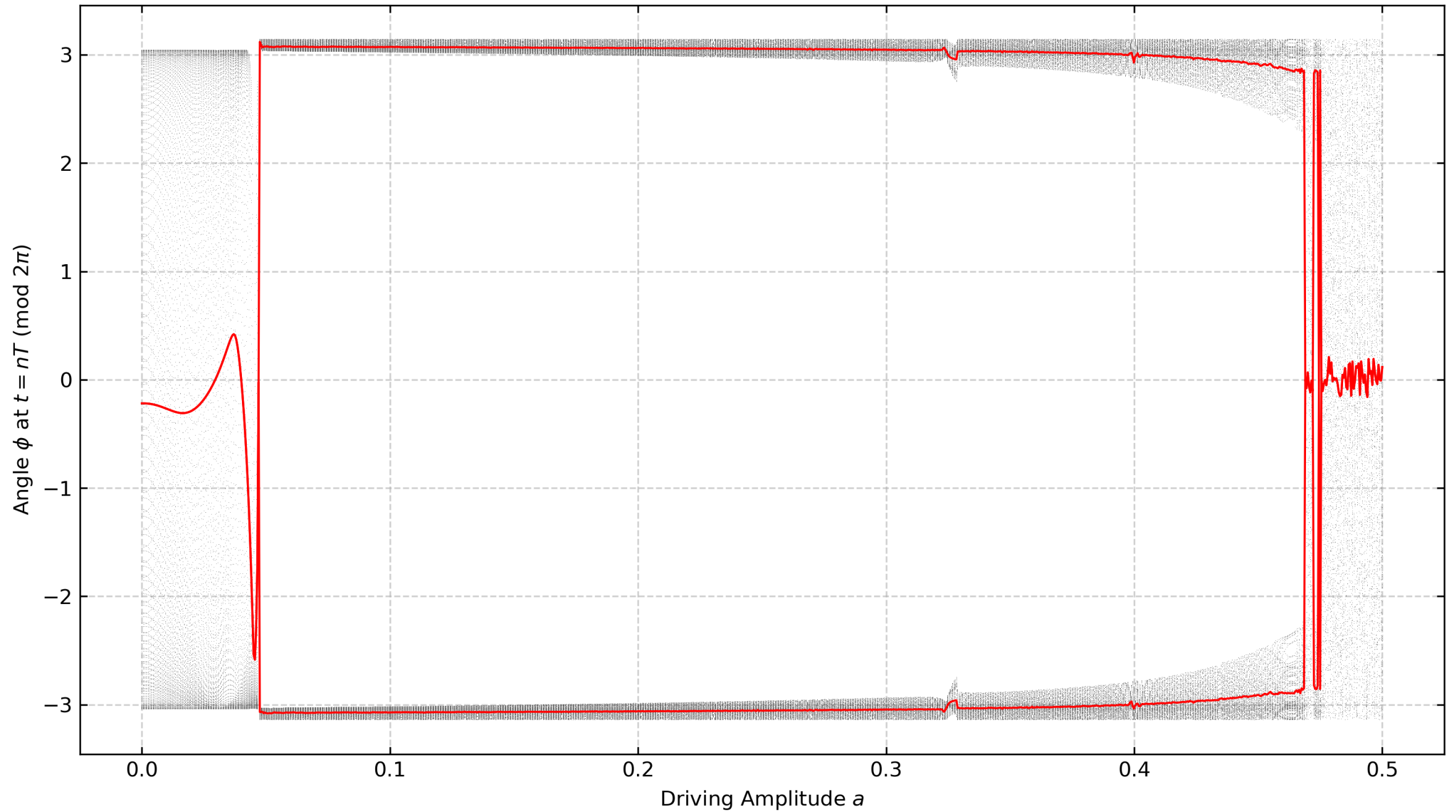
Kapitza Pendulum: $\omega_0 = 1.00$, $\epsilon = 0.25$, $f = 30.00$, $\phi_0 = 3.04$, $\dot{\phi}_0 = 0.00$, $\Delta\phi_0 = 1.0 \times 10^{-6}$, $\Delta\dot{\phi}_0 = 0.00$
 $t = 0.00\text{s}$



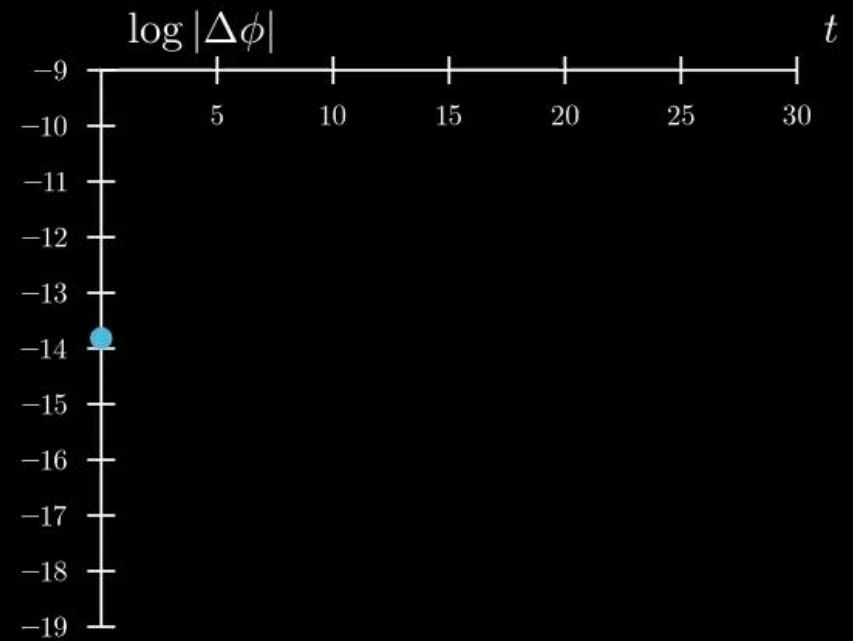
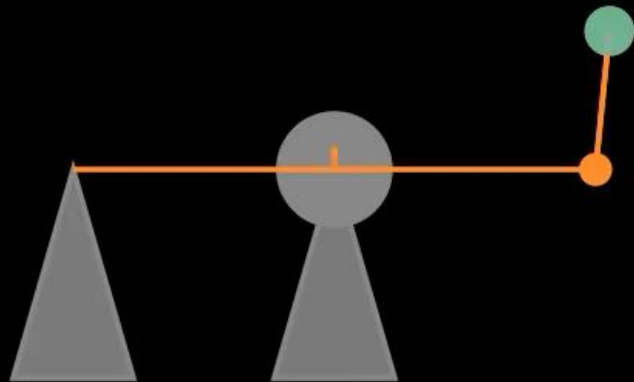
Kapitza Pendulum: $\omega_0 = 1.00$, $\epsilon = 0.50$, $f = 30.00$, $\phi_0 = 3.04$, $\dot{\phi}_0 = 0.00$, $\Delta\phi_0 = 1.0 \times 10^{-6}$, $\Delta\dot{\phi}_0 = 0.00$
 $t = 0.00\text{s}$



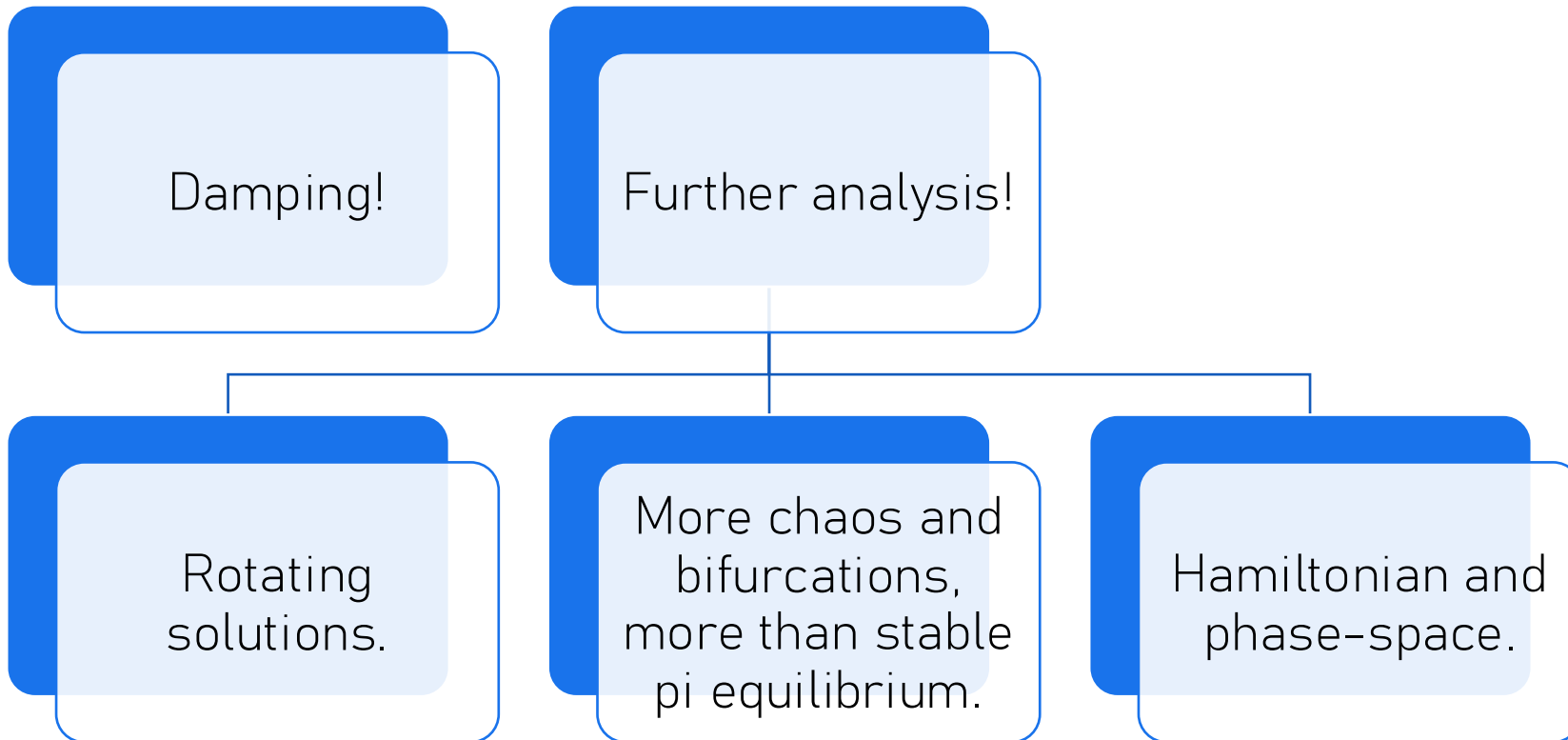
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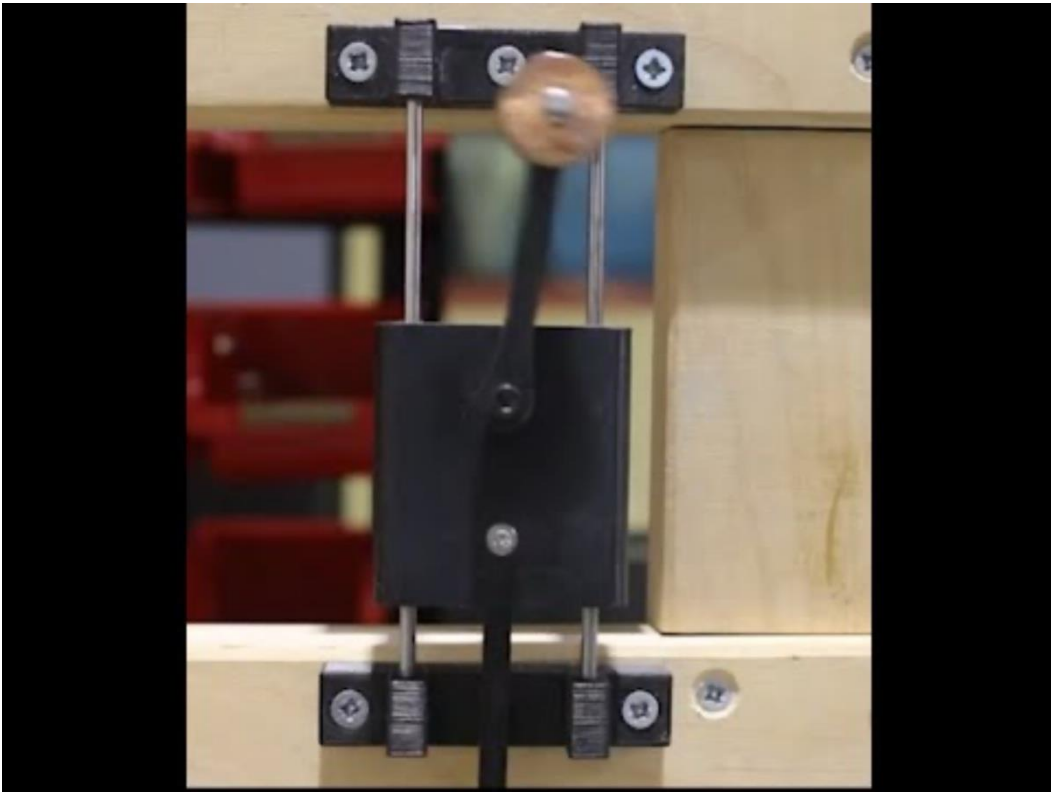
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Further Work



Demonstration



- Video from Dynamics Uci on YouTube.
- https://www.youtube.com/watch?v=GgYABmG_bto

Thank You!

